Network Security and Cryptography

11th November 2024

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

This marking scheme has been prepared as a **guide only** to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Marker's comments:

Moderator's comments:

Mark:	**Moderated mark:**	**Final mark:**

Penalties applied for academic malpractice:

Answer ALL questions

Marks

Question 1

a) A recent vulnerability report identifies that a critical system component is susceptible to a buffer overflow vulnerability: **4**

- Define what a buffer overflow is **and** explain its impact on the system's security and stability.

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Buffer Overflow attacks occur when an application's buffer is overwhelmed with more data than it can handle (1 mark), causing the excess data to overwrite adjacent memory addresses (1 mark).***
- ***Buffer Overflow attacks can lead to system crashes (1 mark) or execution of arbitrary code. (1 mark)***

b) Explain the CIA triad in the context of information security. **6**

Mark Scheme

Award 2 marks for each of the following up to a maximum of 6 marks:

- ***Confidentiality (1 mark): Ensures that information is only accessible to those who are authorised to view it. (1 mark)***

- ***Integrity (1 mark): Ensures that information is accurate and has not been tampered with or altered by unauthorised parties. (1 mark)***

Availability (1 mark): Ensures that information and resources are accessible to authorised users when needed. (1 mark)

Total 10 Marks

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Marks

Question 2

a) Describe **TWO (2)** key differences between Block and Stream Ciphers and give **ONE (1)** example of each cipher. **6**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 6 marks:

- ***Block ciphers encrypt data in fixed-size blocks, typically 64 or 128 bits at a time. (1 mark)***
- ***Block ciphers require padding if the data to be encrypted is not an exact multiple of the block size. (1 mark)***
- ***Block cipher examples: Advanced Encryption Standard (AES) or Data Encryption Standard (DES) or Triple DES (3DES). (1 mark)***
- ***Stream ciphers encrypt data one bit or byte at a time. (1 mark).***
- ***Stream ciphers do not require padding, as they can encrypt data of any length. (1 mark).***
- ***Stream cipher examples: RC4, Salsa20. (1 mark)***

b) Describe **TWO (2)** key differences between symmetric **and** asymmetric cryptography. **4**

Mark Scheme

Award 2 marks for any of the following up to a maximum of 4 marks:

- ***Symmetric Cryptography: Uses a single key for both encryption and decryption (1 mark). Generally faster than asymmetric. (1 mark)***
- ***Asymmetric Cryptography: Uses two different keys: a public key for encryption and a private key for decryption (1 mark). Slower compared to symmetric. (1 mark)***

Total 10 Marks

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Marks

Question 3

a) Explain **THREE (3)** types of attacks that can target symmetric encryption. **6**

Mark Scheme

Award 2 marks for any of the following up to a maximum of 6 marks:

- ***Brute-Force Attack (1 mark): In a brute-force attack, the attacker systematically tries all possible keys until finding the correct one that decrypts the ciphertext to the known plaintext. (1 mark)***
- ***Known-Plaintext Attack (1 mark): In this attack, the attacker possesses both the plaintext and its corresponding ciphertext. The attacker tries to deduce the secret key by analysing the relationship between known plaintext-ciphertext pairs. (1 mark)***
- ***Chosen-Plaintext Attack (1 mark): The attacker can choose specific plaintexts and obtain their corresponding ciphertexts from the encryption oracle. (1 mark)***
- ***Dictionary Attack (1 mark): The attacker attempts a list of common passwords or likely word combinations to crack the ciphertext.***

b) What is password salting **and** explain **THREE (3)** ways it enhances password security.

Mark Scheme

Award up to 1 mark for each of the following up to 4 marks:

- ***Password salting involves adding random data to each password before hashing. (1 mark)***
- ***It prevents attackers from using rainbow tables. (1 mark)***
- ***Slows down or makes brute-force attacks more difficult. (1 mark)***
- ***It makes passwords unique by preventing hash collisions (1 mark)***

Total 10 Marks

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Marks

Question 4

a) Describe **THREE (3)** common security vulnerabilities that can arise from inadequate or improper input validation in software applications. **6**

Mark Scheme

Award 1 mark for any of the following up to a maximum of 6 marks:

- ***SQLi Injection Attacks (1 mark): occurs when user-supplied data is not properly sanitised or parameterised before being included in SQL statements (1 mark).***
- ***Command Injection attacks (1 mark): occurs when user input is passed to functions that execute system commands without proper validation. (1 mark),***
- ***Cross-Site Scripting (XSS) (1 mark): occur when an application includes untrusted data in a web page without proper validation or escaping. (1 mark).***
- ***Buffer Overflow attacks (1 mark): can allow users to input more data than a system's buffer can handle, leading to memory corruption or system crashes. (1 mark).***

b) A user unknowingly visits a fake website and downloads and installs a file that contains malware. Describe **TWO (2)** potential impacts of the malware on the user's computer. **4**

Mark Scheme

Award 1 mark for any of the following to a maximum of 4 marks:

- ***Data Loss and Corruption (1 mark): Malware can delete or corrupt files on a device, leading to the loss of important data. (1 mark)***
- ***Theft of Sensitive Information (1 mark): Malware often aims to steal sensitive information such as login credentials, financial data, and personal details. (1 mark)***
- ***System Damage (1 mark): Some malware can cause serious damage to systems, rendering them inoperable and disrupting normal operations. (1 mark)***
- ***Ransom Demands (1 mark): Ransomware locks or encrypts data, demanding payment for its release. (1 mark)***
- ***Unauthorised Access (1 mark): The malware could allow attackers to gain unauthorised remote access to the user's computer and data. (1 mark)***

Total 10 Marks

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Marks

Question 5

a) Bob has received a confidential email from Alice, encrypted using Pretty Good Privacy (PGP). Describe the steps Bob would take to decrypt the email and verify it was indeed sent by Alice. **4**

Mark scheme

Award 1 mark for each up to a maximum of 4 marks:

- ***Bob uses his private key to decrypt the email. (1 mark)***
- ***Bob checks the digital signature attached to the email from Alice. (1 mark)***
- ***Bob obtains Alice's public key from his PGP keyring or trusted PGP server (1 mark)***
- ***Bob uses Alice's public key to verify this signature matches (1 mark)***

b) Describe what an intrusion detection system (IDS) is **and** explain the difference between signature-based intrusion detection **and** anomaly-based intrusion detection. **6**

Mark Scheme

Award 1 mark for each up to a maximum of 6 marks:

- ***An Intrusion Detection System (IDS) is a security tool (1 mark) designed to monitor network traffic or system activity for malicious behaviour, policy violations, or unauthorised access. (1 marks.***
- ***Signature-Based Intrusion Detection: This method detects intrusions by comparing network traffic or system activity to a database of known attack patterns, called signatures (1 mark). When a match is found, the system raises an alert (1 mark).***
- ***Anomaly-Based Intrusion Detection: This method establishes a baseline of normal network or system behaviour and then detects deviations from this norm (1 mark). Any unusual activity is flagged as potentially malicious (1 mark)***

Total 10 Marks

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Marks

Question 6

a) Explain what a Virtual Private Network (VPN) is **and** describe the **TWO (2)** primary types of VPNs. **6**

Mark scheme

Award 1 mark for each up to a maximum of 6 marks:

- ***VPN, or Virtual Private Network, is a technology that creates a secure, encrypted connection (1 mark) over an untrusted public network, typically the internet. (1 mark)***
- ***Site-to-Site VPN (1 mark): connects entire networks, such as linking the networks of two different office locations. It allows for secure communication between multiple sites. (1 mark)***
- ***Remote Access VPN (User-to-Site) (1 mark): allows individual users to connect securely to a network from a remote location. (1 mark)***

b) Describe **TWO (2)** common Virtual Private Network (VPN) protocols. **4**

Mark scheme

Award 1 mark for any of the following up to a maximum of 4 marks:

- ***IPsec (Internet Protocol Security) (1 mark): is a comprehensive suite of protocols designed to secure Internet Protocol (IP) communications, allowing for the secure transmission of data over the Internet. (1 mark)***
- ***PPTP (Point-to-Point Tunnelling Protocol) (1 mark): is a tunnelling protocol used to implement virtual private networks (VPNs). (1 mark)***
- ***L2TP (Layer 2 Tunneling Protocol) (1 mark): is a VPN protocol that creates a secure tunnel for data transmission but doesn't provide encryption on its own, typically requiring IPSec for security. (1 mark)***
- ***SSTP (Secure Socket Tunneling Protocol) (1 mark): is a Microsoft-developed VPN protocol that uses SSL/TLS encryption to create a secure connection and can easily bypass firewalls. (1 mark)***
- ***WireGuard (1 mark): is a modern, lightweight VPN protocol designed for enhanced security and efficiency. (1 mark)***

Total 10 Marks

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Marks

Question 7

a) You are tasked with securing a small business's Wi-Fi network. Describe **THREE (3)** essential steps you would take to enhance the security of the network. **6**

Mark scheme

Award 1 mark for each of the following up to a maximum of 6 marks:

- ***Use WPA2 Encryption (1 mark): Employ WPA2 with a strong, unique passphrase for network encryption (1 mark).***
- ***Change Default Settings (1 mark): Modify default settings on routers and access points, including default usernames, passwords, and SSIDs, to reduce vulnerability. (1 mark):***
- ***Enable MAC Filtering (1 mark): Use MAC address filtering to restrict access only to devices with authorised MAC addresses. (1 mark)***

b) As a security analyst at a financial institution implementing a new online banking system, describe the **TWO (2)** primary types of risk assessments you could use to evaluate potential risks. **4**

Mark scheme

Award 1 mark for each of following up to a maximum of 4 marks:

- ***Qualitative Risk Assessment (1 mark): Involves subjective evaluation of risks, often using descriptive terms like low, medium, or high. (1 mark)***
- ***Quantitative Risk Assessment (1 mark): Involves numerical analysis, assigning monetary values to risks. (1 mark)***

Total 10 Marks

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Marks

Question 8

a) Explain what a hash function is **and** describe **TWO (2)** main uses. **6**

Mark scheme

Award 1 mark for any of the following up to a maximum of 6 marks:

- ***A hash function is a mathematical algorithm that transforms input data or a message (1 mark) into a fixed-size string of characters known as a digest (1 mark).***
- ***It is a one-way process (can't decrypt) (1)***
- ***Data Integrity Verification (1 mark): Hash functions are used to verify that data has not been altered (1 mark).***
- ***Password Storage (1 mark): Hash functions securely store passwords by converting them into hash values (1 mark).***

b) Explain **TWO (2)** examples of biometric authentication methods. **4**

Mark scheme

Award 2 marks for any of the following up to a maximum of 4 marks:

- ***Fingerprint Recognition (1 mark): This method analyses and matches the unique patterns of ridges and valleys on a person's fingerprint. (1 mark)***
- ***Facial Recognition (1 mark): Facial recognition systems map the unique features of an individual's face (1 mark).***
- ***Iris or Retinal Scanning (1 mark): These methods analyse the intricate patterns in a person's iris or retina. (1 mark)***
- ***Voice Recognition (1 mark): This technique identifies individuals by analysing their unique vocal characteristics, including pitch, tone, and speaking style. (1 mark)***

Total 10 marks

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Marks

Question 9

a) Describe the difference between hardware **and** software firewalls. **4**

Mark scheme

Award 1 mark for each up to a maximum of 4 marks:

- ***Software firewalls are located on individual computers and protect only that computer;(1 mark). Software firewalls are usually easy to install and come with an easy-to-use interface that allows a user to configure them in a highly flexible way. (1 mark)***

Hardware firewalls typically reside in routers and protect the whole network or part of a network. (1 mark). Hardware firewalls protect a whole network or a section of a network by setting up network-wide traffic filters. (1 mark)

b) Describe **THREE (3**) weaknesses of biometric authentication. **6**

Mark Scheme

Award 1 mark for each up to a maximum of 6 marks:

- ***Privacy Concerns (1 mark): Collecting and storing biometric data raises privacy concerns. (1 mark)***
- ***Cost (1 mark): Implementing biometric systems can be expensive due to the need for specialised hardware and software. (1 mark)***

False Positives/Negatives (1 mark): Biometric systems may produce false positives (incorrectly accepting an unauthorised user) or false negatives (rejecting an authorised user). (1 mark)

Total 10 marks

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Marks

Question 10

a) Explain why WPA2 is considered more suitable than WPA for securing wireless networks. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***WPA2 uses the stronger AES encryption algorithm. (1 mark)***
- ***WPA (Wi-Fi Protected Access): Susceptible to brute-force or dictionary attacks on pre-shared keys due to the use of TKIP (1 mark)***
- ***WPA2 generally offers better performance compared to WPA. (1 mark)***
- ***WPA2 is compatible with newer devices compared to WPA. (1 mark)***

b) Explain the concept of zero trust **and** identify **TWO (2)** key principles of zero trust. **6**

Mark Scheme

Award 2 mark for definition:

- ***Zero trust is a security model (1 mark) that assumes no user, device, or system, whether inside or outside the network can be trusted by default (1 mark).***

Award 2 marks for any of the following to a maximum of 4 marks:

- ***Never Trust, Always Verify (1 mark): Verify the identity of all users and devices, even those within the network (1 mark).***
- ***Least Privilege (1 mark): Grant the minimum level of access required for tasks (1 mark).***
- ***Micro-Segmentation (1 mark): Isolate workloads and data to limit lateral movement (1 mark).***

Total 10 Marks

End of Paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1,3	Yes
2	1,2,3	Yes
3	1,2,3	Yes
4	3	Yes
5	4	Yes
6	4	Yes
7	1,5,6	Yes
8	2.3	Yes
9	3,4	Yes
10	1,5,6	Yes

Grade descriptors

Learning Outcome	Pass	Merit	Distinction
Understand the basic concept and application of Network Security and Cryptography	Demonstrate adequate understanding and application of Network Security and Cryptography	Demonstrate robust understanding and application of Network Security and Cryptography algorithm	Demonstrate a highly comprehensive understanding and application of Network Security and Cryptography.
Critically evaluate the suitability of different Cryptographic Algorithms	Demonstrates a basic evaluation of different cryptographic algorithms, identifying some strengths and weaknesses.	Demonstrates a thorough evaluation of different cryptographic algorithms, with a detailed analysis of their strengths and weaknesses, and appropriate comparisons.	Demonstrates a comprehensive and critical evaluation of different cryptographic algorithms, with in-depth analysis, insightful comparisons, and a well-argued justification of their suitability in various contexts.
Understand the vulnerabilities and systematically evaluate the security risks in networks	Demonstrates a basic understanding of vulnerabilities and can evaluate security risks in networks with guidance.	Demonstrates a good understanding of vulnerabilities and can systematically evaluate security risks in networks with some analysis.	Demonstrates a comprehensive understanding of vulnerabilities and can critically and systematically evaluate security risks in networks with detailed analysis.

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Critically analyse appropriate tools and techniques for network security	Demonstrates a basic ability to analyse appropriate tools and techniques for network security, with limited critical evaluation.	Demonstrates a good ability to analyse appropriate tools and techniques for network security, with a clear and reasoned critical evaluation.	Demonstrates an excellent ability to critically analyse appropriate tools and techniques for network security, providing comprehensive and insightful evaluation.
Understand the wireless network security and associated protocols	Demonstrates a basic understanding of wireless network security and its associated protocols.	Demonstrates a good understanding of wireless network security and its associated protocols, with clear explanations and some examples.	Demonstrates a comprehensive understanding of wireless network security and its associated protocols, providing detailed explanations, examples, and critical insights.
Explain the concepts of Information Security and explain good practices in achieving security.	Demonstrates a basic understanding of the concepts of Information Security and provides examples of good practice in achieving security.	Demonstrates a clear and detailed understanding of the concepts of Information Security, explaining them accurately and providing well-chosen examples of good practices in achieving security.	Demonstrates a comprehensive and insightful understanding of the concepts of Information Security, explaining them thoroughly and providing detailed, relevant examples of best practices in achieving security.

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